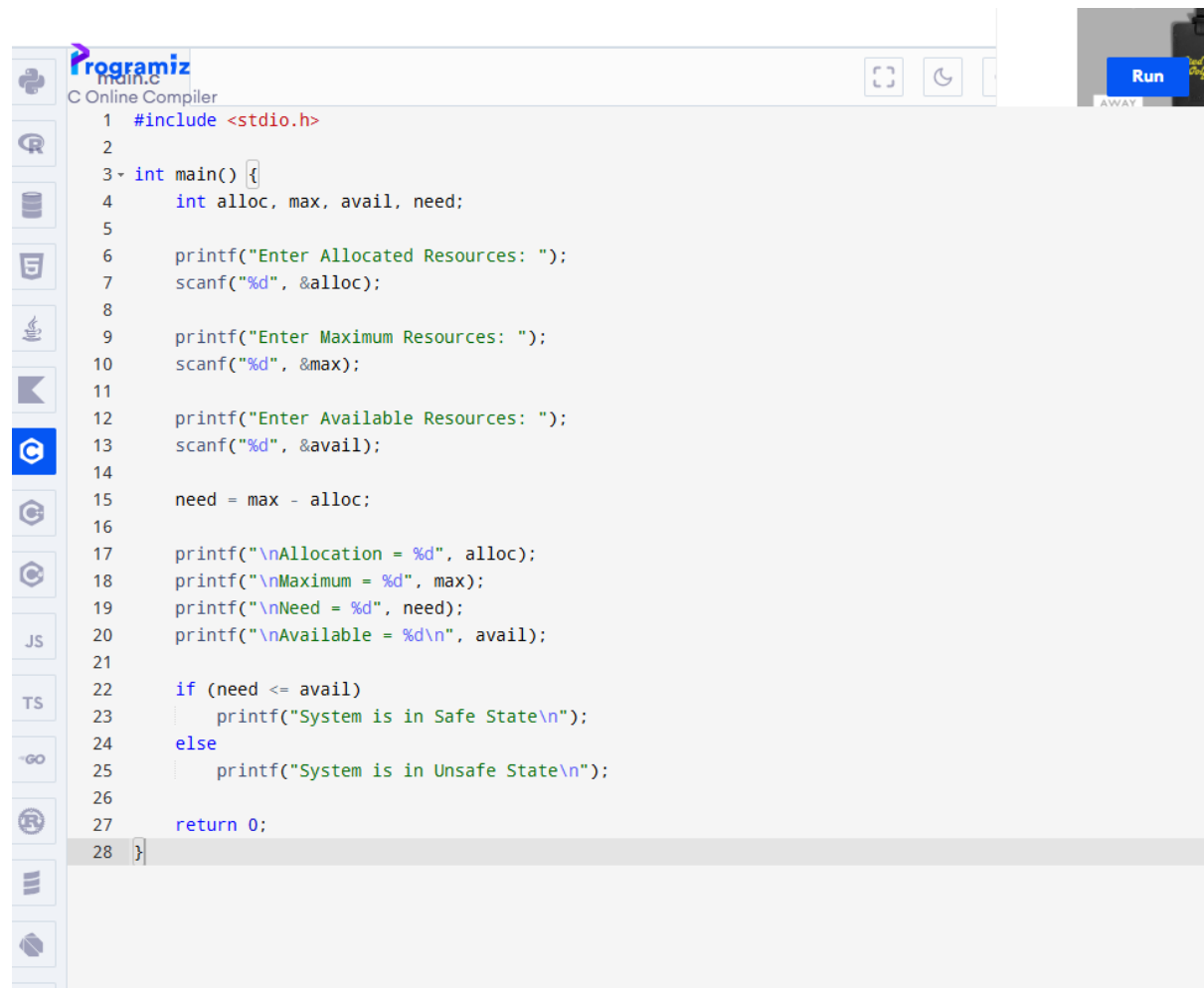


17. Illustrate the deadlock avoidance concept by simulating Banker's algorithm with C.

Code:



```
1  #include <stdio.h>
2
3  int main() {
4      int alloc, max, avail, need;
5
6      printf("Enter Allocated Resources: ");
7      scanf("%d", &alloc);
8
9      printf("Enter Maximum Resources: ");
10     scanf("%d", &max);
11
12     printf("Enter Available Resources: ");
13     scanf("%d", &avail);
14
15     need = max - alloc;
16
17     printf("\nAllocation = %d", alloc);
18     printf("\nMaximum = %d", max);
19     printf("\nNeed = %d", need);
20     printf("\nAvailable = %d\n", avail);
21
22     if (need <= avail)
23         printf("System is in Safe State\n");
24     else
25         printf("System is in Unsafe State\n");
26
27     return 0;
28 }
```

Output:



```
Output

Enter Allocated Resources: 3
Enter Maximum Resources: 7
Enter Available Resources: 5

Allocation = 3
Maximum = 7
Need = 4
Available = 5
System is in Safe State

=== Code Execution Successful ===
```